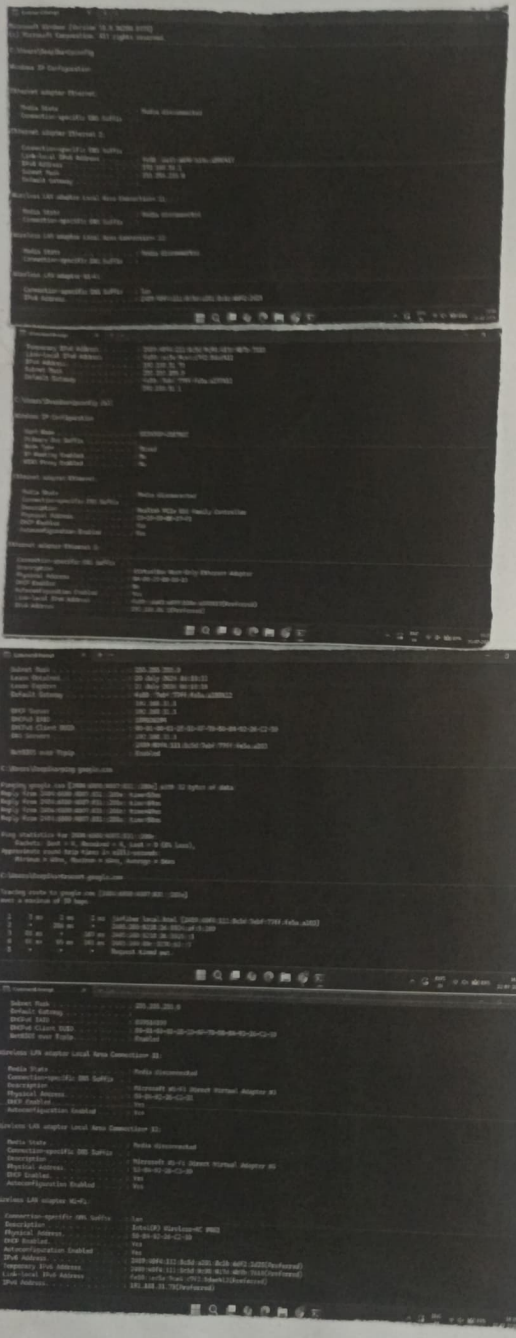


EXP NO: 1

DATE: 16/07/26

BASIC NETWORKING COMMANDS IN LINUX AND WINDOWS OPERATING SYSTEMS

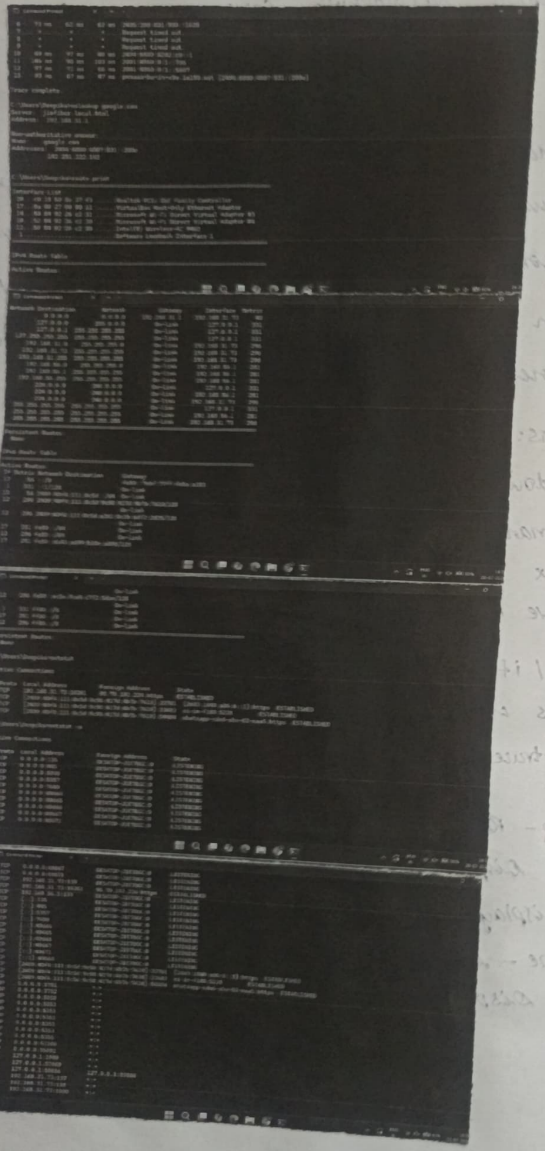
AIM:

To study and execute basic networking commands in windows and Linux operating systems for viewing network configuration, testing connectivity, checking routing information, monitoring network status and troubleshooting network-related issues.

REQUIREMENTS:

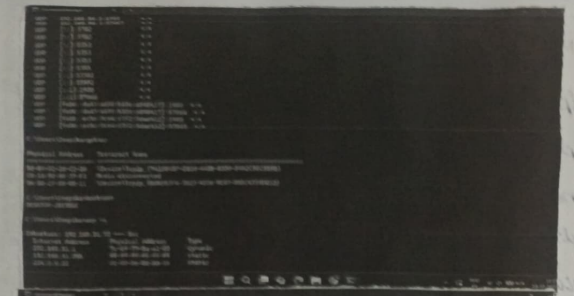
- Windows 10/11 or Linux (Ubuntu 20.04/22.04)
- Command prompt(window)
- Linux
- Active network (LAN/Wifi)

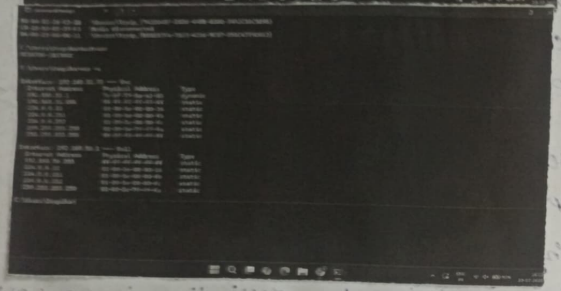
- ip config / if config /ip - displays network configuration
- Ping - tests connectivity to a remote host
- tracert / traceroute - Displays the path packets take to reach a destination.
- nslookup - Revolves domain name to Ip address
- netstat - Displays active network connection ports.
- arp - Displays ARP cache
- hostname - Displays the system hostname.
- route - Displays or modifies the routing table

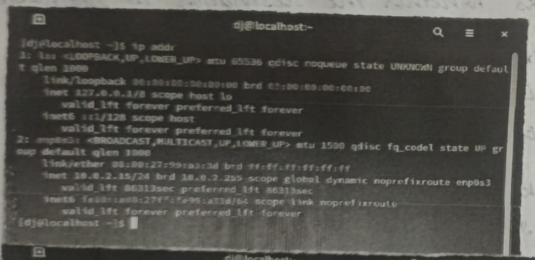


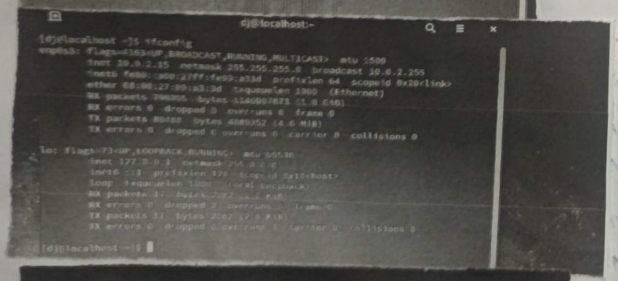
WINDOWS COMMANDS:

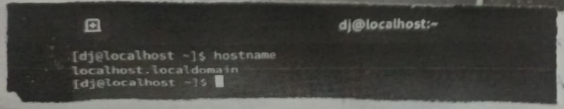
- Command : ipconfig
  Displays : Ip address
  Subnet Mask
  Default gateway
- Command : ipconfig /all
  Displays : MAC address
  DHCP Server
  DNS Server
  Lease Information
  Adapter Details
- Command : ping google.com
  Displays : checks whether the destination is reachable.
- Command : tracert google.com
  Displays : Displays every router (hop) between your computer and Hardware.
- Command : nslookup google.com
  Displays : DNS server, Ip address of website
- Command : route print
  Displays : Network routes, Gateway, Destination Networks
- Command : netstat
  Displays : Active Tcp connections, Listening ports
- Command : netstat -a
  Displays : All ports, Tcp connections, Udp ports.
- Command : getmac
  Displays : Displays MAC address of the computer.







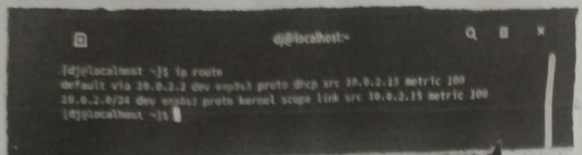


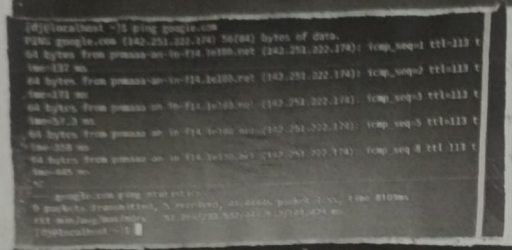


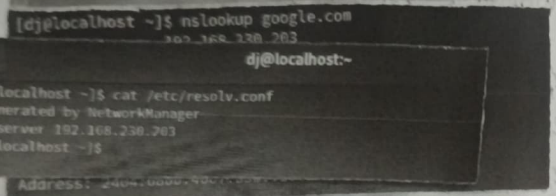
- Command: hostname
  Purpose: displays computer name
- Command: arp -a
  Purpose: Displays Ip-to-MAC address mappings
- Command: netstat -a
  Purpose: All ports, Tcp connections, UDP ports

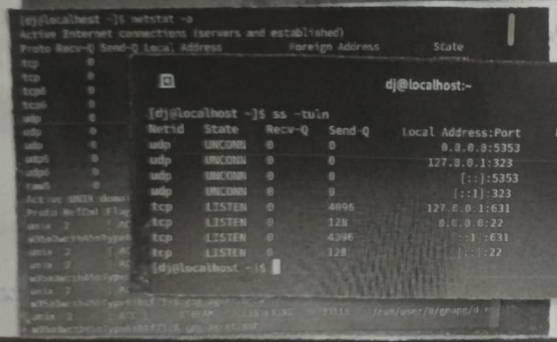
LINUX NETWORKING COMMANDS:

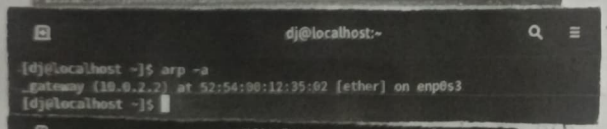
- Command: ip addr or ifconfig
  Purpose: Displays Ip address
- Command: host name
  Purpose: displays system hostname
- Command: ip route
  Purpose: shows default Gateway, Routing table
- Command: ping google.com
- Command: traceroute google.com
  (if not installed): sudo apt install traceroute
- Command: netstat -a (or) ss -a
  Purpose: displays network statistics
- Command: arp -a (or) ip neigh
  Purpose: displays ARP table
- Command: ip link
  Purpose: displays MAC address
- Command: cat /etc/ resolv.conf
  Purpose: DNS configuration.

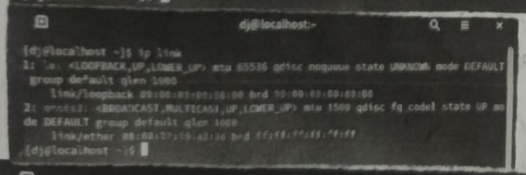


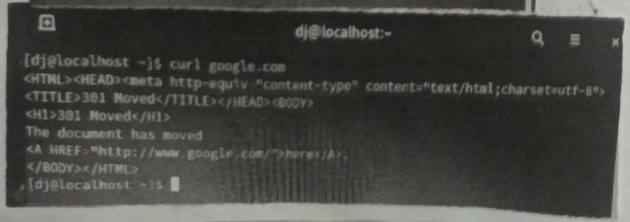












- command: ip link show
  displays: • UP / Down status
  • Interface name
  purpose: check interface status
- command: ss -tuln
  purpose: displays • Tcp ports
  • Udp ports
  • Listening services
- command: curl google.com (or) wget google.com
  purpose: To check internet connectivity.

RESULT:

The basic networking commands in Linux & Windows O.S has been implemented & executed successfully